Working with Gateway



CC by 4.0

WSO2

Objective

At the end of this module, attendees will get a basic idea of

- What is the API Gateway?
- API Gateway request flow
- Gateway Environments
- Gateway Caching

Understanding API Gateway

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API Gateway

- API Gateway is the runtime component of API Manager and acts as a proxy for API invocations.
- When an API is published from the Publisher Portal, it is exposed to the users via API Gateway.
- Gateway directs the API calls to the backend endpoints and enforces security, rate limiting, does stats publishing, etc.
- WSO2 API Manager Gateway is developed based on the WSO2 EI, which is known for its high performance.
- The default Gateway ports

  HTTPS:- 8243

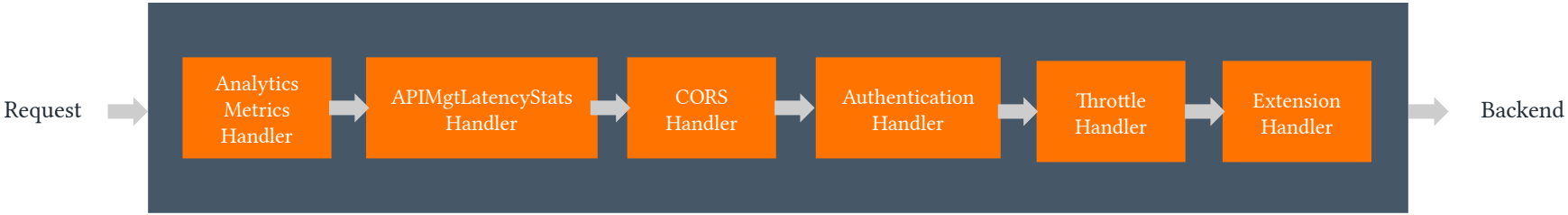
  HTTP:- 8280

API Gateway Request/ Response Flow

Understanding Handlers

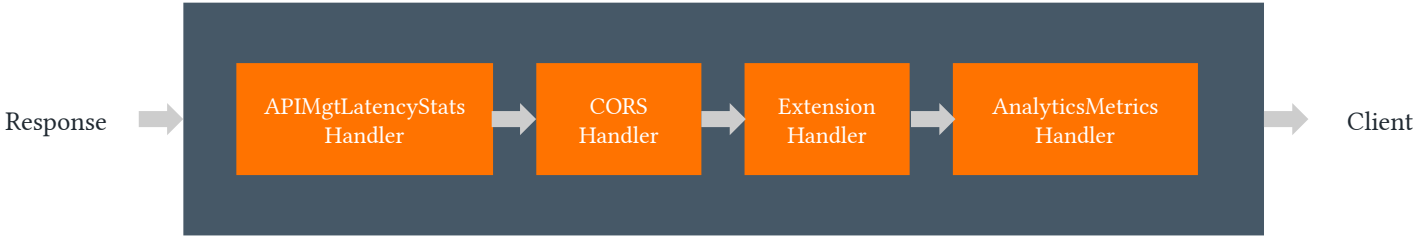
- In API Gateway, the request and response flow are controlled through a sequence of handlers.
- Handler is a synapse component which can intercept a request and response and perform a specific task on it.
- Gateway use handlers for,
    - Authentication
    - Throttling
    - CORS Handling
- Gateway uses a Global Handler for,
    - Analytics Publishing
- Based on the requirement additional handlers can be implemented and integrated.

Request Flow (IN Flow)



Analytics Metrics Handler	Capture the client information for analytics publishing.This is the global synapse handler to publish analytics data to analytics cloud. (user agent, IP etc)
APIMgtLatencyStats Handler	Initiates the analytics data collection. (request start time)
CORS Handler	Validates the enforced Cross Origin Resource Sharing conditions.
Authentication Handler	Does the token validation. If the token is opaque, it will be validated through key manager. If the token is self contained JWT token, API Key or Mutual TLS, it will be validated in the gateway it self.
Throttle Handler	Publish request information to traffic manager
Extension Handler	Executes any custom mediation sequences applied in the request flow. (IN Flow)

Response Flow (OUT Flow)



APIMgtLatencyStats Handler	Collects the time the response is received
CORS Handler	Execute any CORS policies applied in response path
Extension Handler	Executes any custom mediation sequences applied in the response flow. (OUT Flow)
AnalyticsMetrics Handler	Publishes the captured stats to the analytics server.

Changing Default API Flows

Operation Policies

- Policies generally enforce some business logic that needs to be executed on the request, response, or fault flow of an API invocation.

  IN Flow : The API request path from Client to Backend

  OUT Flow : API response path. Backend to Client

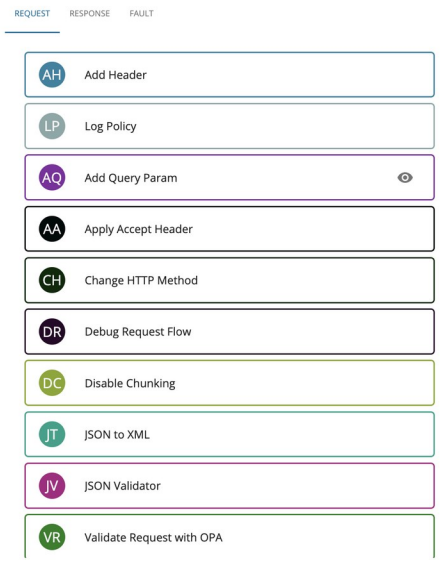
  Fault Flow: If any error occurred during the request/response flow

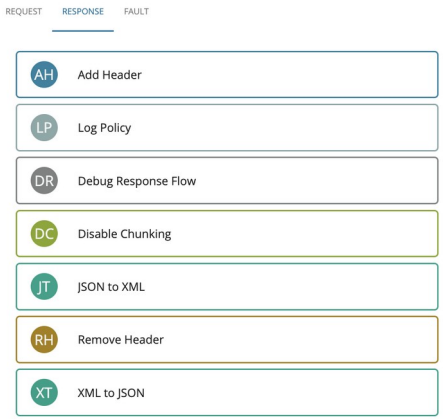
Default Operation Policies

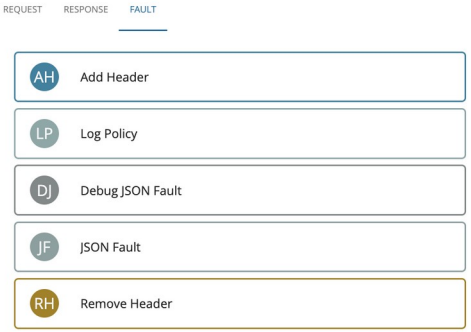
- API Manager ships a default set of policies that cover most of the common use cases that you will need.

  Add Header : Set a header in the request/response/fault flow.
  Remove Header : Remove a header in the request/response/fault flow.
  Add Query Param : Add a query param in the request flow.
  Change HTTP Method : Change the HTTP method of a resource in the request flow.
  Rewrite Resource Path : Rewrite the resource path in the request flow.

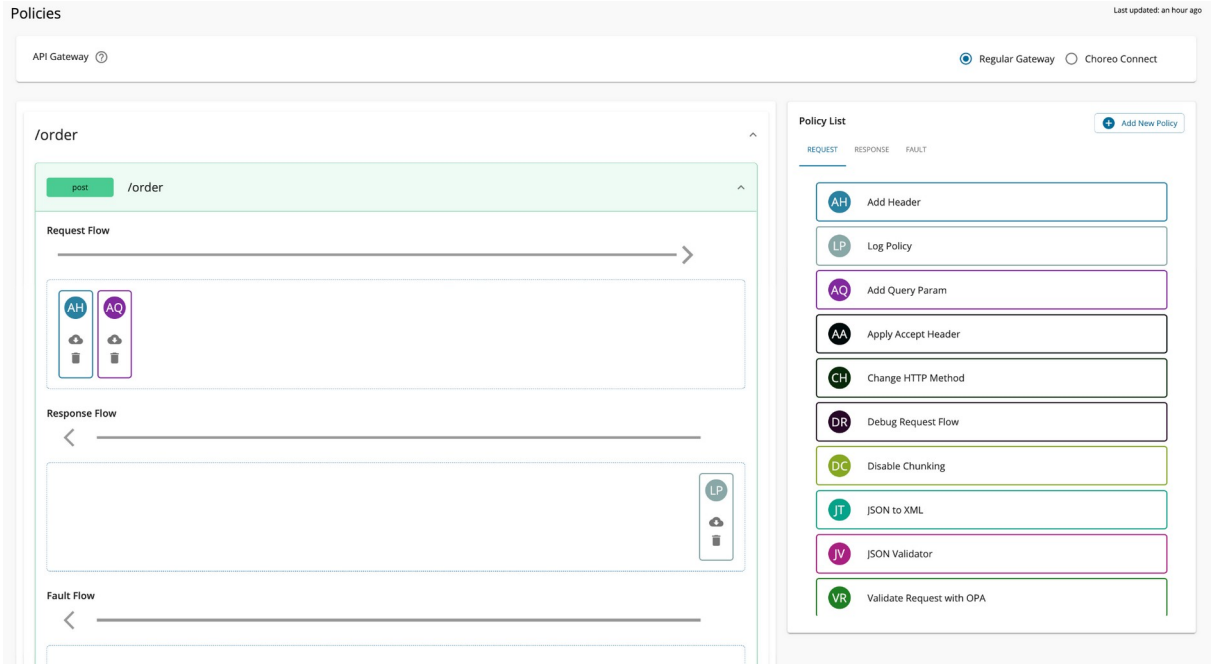
Default Operation Policies







Attaching API Policies to Resources



Gateway Environments

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Gateway Environments

- API Gateways are considered as Environments.
- Each environment has environment name and address
- Environments are defined in <APIM_HOME>/repository/conf/deployment.toml file.

```
[[apim.gateway.environment]]

name = "default"

type = "hybrid"

service_url = "https://localhost:${mgt.transport.https.port}/services/"

username= "${admin.username}"

password= "${admin.password}"

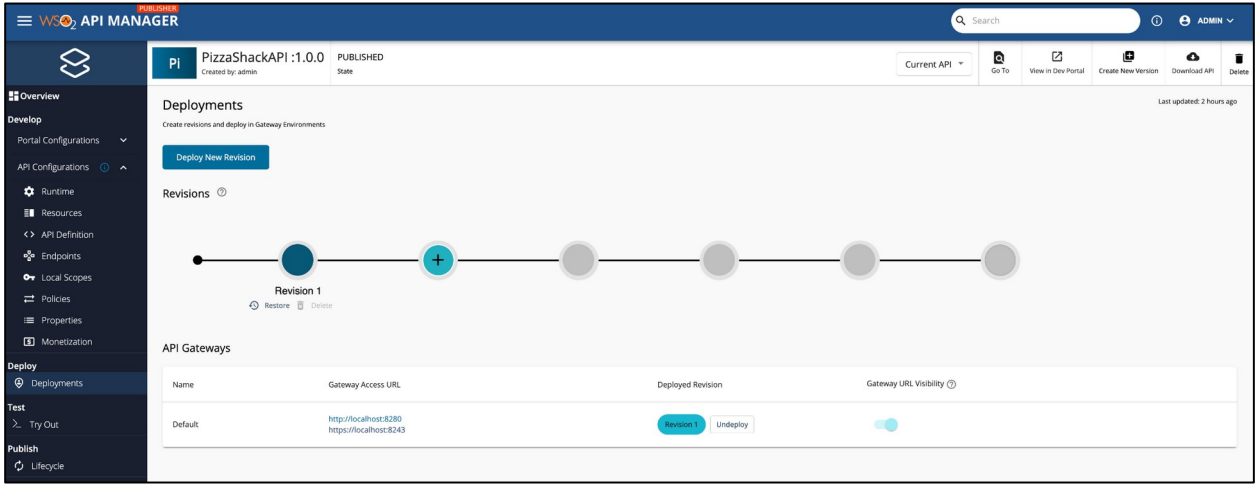
http_endpoint = "http://localhost:${http.nio.port}"

https_endpoint = "https://localhost:${https.nio.port}"
```

- In the API Publisher portal the registered Gateway environments are listed. API Publisher can select required environments when publishing the API.



Select Gateway Environment in Publisher



- In API Publisher, open an API by selecting it.
- Go to Deployments tab in left menu.
- Select the revision you want to deploy to the gateway.
- Click on deploy to save the changes.

If the API is in created state, go to the Lifecycle page and click on Publish button to publish the API to the API developer portal and make it discoverable to subscribers.

Gateway Federation

- From version 4.5.0 APIM supports deploying APIs to external third-party Gateways.
- In this scenario APIM acts as the centralized control plane where the users create, design and govern the APIs deployed on WSO2 and external runtimes such as AWS, Azure, Kong, and Envoy.
- WSO2 provides inbuilt support for these federated gateways, and users can add other gateways by configuring custom gateway connectors.

Federated Gateways

- Supported Federated Gateways

  AWS API Gateway – deploy & discover APIs on AWS

  Azure API Management – deploy & discover APIs in Azure, Entra ID integration

  Kong Gateway – discover APIs and services (Kubernetes & Standalone)

  Envoy Gateway – discover APIs in Kubernetes environments

- Gateway Modes

  Write-Only Mode - deploy APIs from APIM to external gateway

  Read-Only Mode - discover and import existing APIs into APIM

  Read-Write Mode - Providing bidirectional synchronization (deploy + discover)

Caching

Gateway Caching - Introduction

- When a request is passing through a gateway, it performs several functions, such as security checks etc.
- In order to make these processes faster, Gateway uses several types of caches.

  Gateway Token Cache

  Resource Cache

  Response Cache
- Default cache expiry time is 15 minutes and can be configured in <APIM_HOME>/repository/conf/deployment.toml file.

Configure Caching

Artifact Synchronization

Artifact Synchronization

- Maintain consistency among the nodes using one of the below approaches.
- Inbuilt Artifact Synchronizer

  Based on a new gateway REST API which is capable of notifying the changes in the artifacts to the respective nodes in the deployment

  An extension point needs to be configured in the Publisher profile to store the Synapse artifacts in a persistent storage

  When an API is published/edited/removed an event will be sent to the Traffic Manager using Event Notifiers with the API Name, UUID, and the Gateway label of the API

  Gateways are subscribed to the Traffic Manager.

  Upon receiving an event, extension in the Gateway gets the Synapse artifacts and deploy them in the memory